1 Injections — IM / SC / ID

HME922 Procedural Station · 10 minutes · Injections guideline + IM/SC DOPS form

guideline wording — say it aloud DOPS On the examiner's DOPS marking form ☐ tick when done

KEY MESSAGE

Right route, right site, right needle for the drug **and** the patient's body habitus. Check the **6 Rights**. IM: aspirate only by clinical judgement. SC: **never aspirate**.

UNIVERSAL OPENING

- ☐ Hand hygiene — Moment 1 DOPS
- ☐ Introduce yourself: name + "medical student"
- ☐ Confirm ID — **3 identifiers: full name, DOB, UR number/address**; check band + medication chart
- ☐ Explain procedure + why an injection (not oral) in plain language
- ☐ Ask about **allergies** (drug, latex, tape, skin prep) DOPS
- ☐ Allow questions DOPS
- ☐ Gain **informed verbal consent** DOPS
- ☐ Privacy, lighting, exposure, bed height up DOPS

MEMORY AID — 6 RIGHTS

Patient · **D**rug · **D**ose · **R**oute · **T**ime · **D**ocumentation — "Please Don't Disturb Rob's Tea, Doc"

5 Moments of Hand Hygiene: 1 before touching patient · 2 before procedure · 3 after body-fluid risk · 4 after touching patient · 5 after touching surroundings

CLINICAL KNOWLEDGE — SAY IT

- ☐ **Why inject:** drug not absorbed in GIT or too variable; route set by drug (e.g. lipid solubility) + manufacturer
- ☐ **Body habitus:** thick fat + short needle = accidental SC instead of IM
- ☐ **Check medication chart:** right drug, dose, patient, route, time DOPS

Complication	Prevent / manage
Local irritation, allergy, anaphylaxis	Check allergies to every ingredient
Infection	Hand hygiene, skin prep, infection control
Pain	Positioning + technique
Haematoma, bruising (anticoagulants)	Careful site, positioning, technique
Scar tissue / lipohypertrophy	Rotate sites (e.g. insulin)
Vessel injury	Know anatomy; aspirate if indicated
Nerve injury (esp. **sciatic**)	Landmark technique
Vaccine reaction	Observe at least **15 minutes**

CHOOSE NEEDLE & SITE

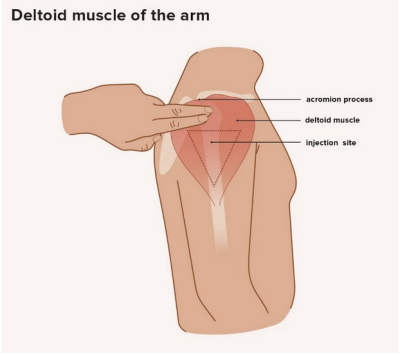
Route	Adult sites	Length	Gauge	Angle	Volume
IM	**Deltoid** · ventrogluteal (best if cachectic) · **vastus lateralis** · dorsogluteal (avoid if obese)	25–40 mm (76 mm large)	21–25 G	**90°**	1–5 mL syringe
SC	**Abdomen** (not within 50 mm of umbilicus) · thigh · upper outer triceps · upper buttock	Insulin 4–13 mm; other 13–16 mm	23/25 G	**45–90°**	≤ 1.5 mL (3 mL abdo)
ID (intradermal)	Anterolateral forearm · upper chest · upper back	10–19 mm	23/25 G	**5–15°** bevel up	0.1 mL (max 0.5)

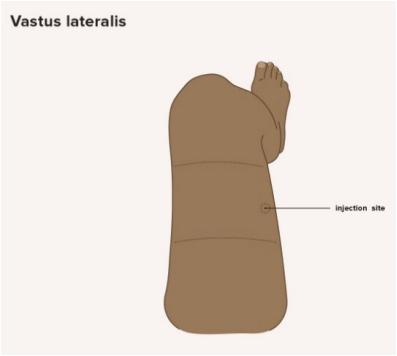
MEMORY AID — ANGLES

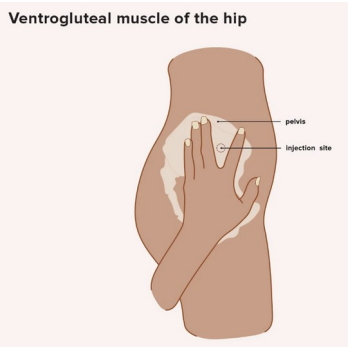
Route	Adult sites	Length	Gauge	Angle	Volume

Intra**M**uscular = **90** · **S**ub**C**ut = **45–90** · intra**D**ermal = **D**ead flat (5–15)
Getting deeper = getting steeper.

IM LANDMARKS



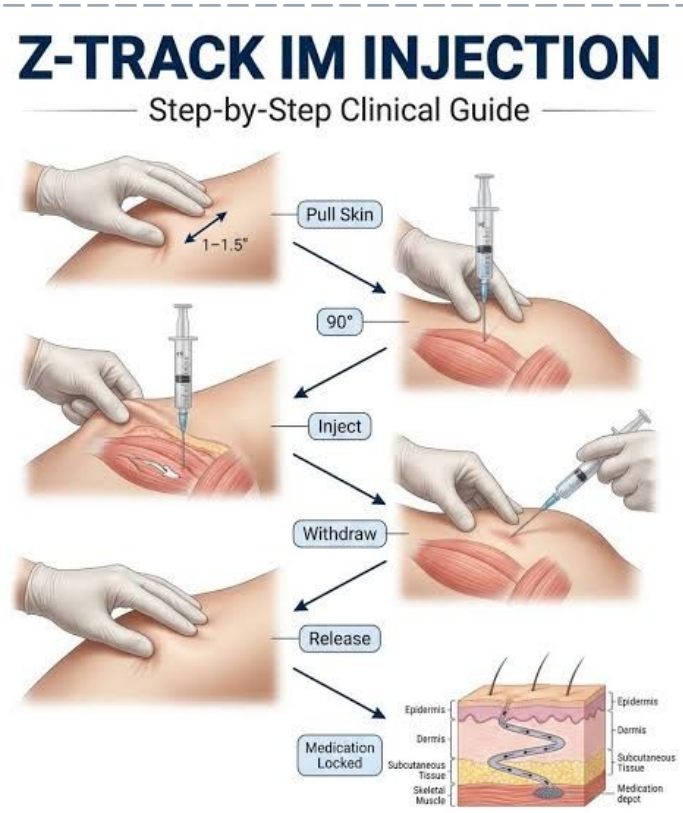




- ☐ **Deltoid:** sitting or hand on hip; expose whole upper arm; acromion + deltoid tuberosity; inverted triangle 2–3 fingers below acromion; inject centre
- ☐ **Vastus lateralis:** greater trochanter → lateral condyle; middle third, anterolateral thigh (rectus femoris = mid-anterior)
- ☐ **Ventrogluteal:** on side; heel of OPPOSITE hand on greater trochanter, index finger on ASIS, spread middle finger → "V"; inject middle
- ☐ **Dorsogluteal:** upper outer quadrant, 5–7.5 cm below iliac crest (crest, PSIS, ischial tuberosity, greater trochanter) — sciatic risk

PROCEDURE — IM (Z-TRACK)

1. Gather: 21–25 G needle, 1–5 mL syringe, swab, cotton ball/dressing, sharps bin
2. Hand hygiene + PPE; consent; 6 Rights
3. Locate site using landmarks DOPS
4. Swab site and **allow to dry** DOPS
5. Non-dominant hand pulls skin **2–3 cm sideways** (Z-track)
6. Dart needle in quickly and smoothly at **90°** DOPS
7. Aspiration **no longer routinely recommended** — clinical judgement by site; follow local policy
8. Inject smoothly at the drug's recommended rate
9. Withdraw needle **then** release skin (seals the track)
10. Cotton ball / dressing; needle to sharps DOPS



PROCEDURE — SC

1. 23/25 G needle, 1–3 mL syringe; select site and **rotate** sites
2. Swab, allow to dry
3. **No pinch** (short needle / enough fat) → 90°
4. **Pinch** (lift skin + fat off muscle) → 45–90°; 45° if cachectic
5. **DO NOT ASPIRATE**
6. Inject **slowly**; withdraw; cotton ball

PROCEDURE — ID

1. 1 mL syringe, bevel **up**, 5–15°
2. Insert only until bevel is covered
3. Inject slowly until a **bleb / wheal** forms
4. Uses: local anaesthetic, allergy testing, tuberculin (Mantoux), some vaccines

UNIVERSAL CLOSING

- ☐ Needle straight into sharps — **never recap**
- ☐ Waste: sharps / biohazard / general DOPS
- ☐ Area clean; patient comfortable; bed lowest; call bell DOPS
- ☐ Aftercare: report pain, swelling, rash, breathing difficulty (vaccines: wait 15 min)
- ☐ Hand hygiene — Moments 4/5 DOPS
- ☐ Document date, time, drug, dose, route, site, needle size, batch (vaccines), name + signature DOPS
- ☐ Can't answer? "I'll pass that on to someone senior"
- ☐ Examiner asks **"How do you think that went?"** → one thing done well · one to improve · one strategy

MY NOTES